



DeepLearning.AI

AI Prompting for Everyone

Working with
Multimedia & Code



DeepLearning.AI

Working with Multimedia & Code

Working with
multimedia

AI models can generate images, videos, voices, music, code, and more

A delicious cake that came to life



Generated with Nano Banana

Man shrinking to bug size



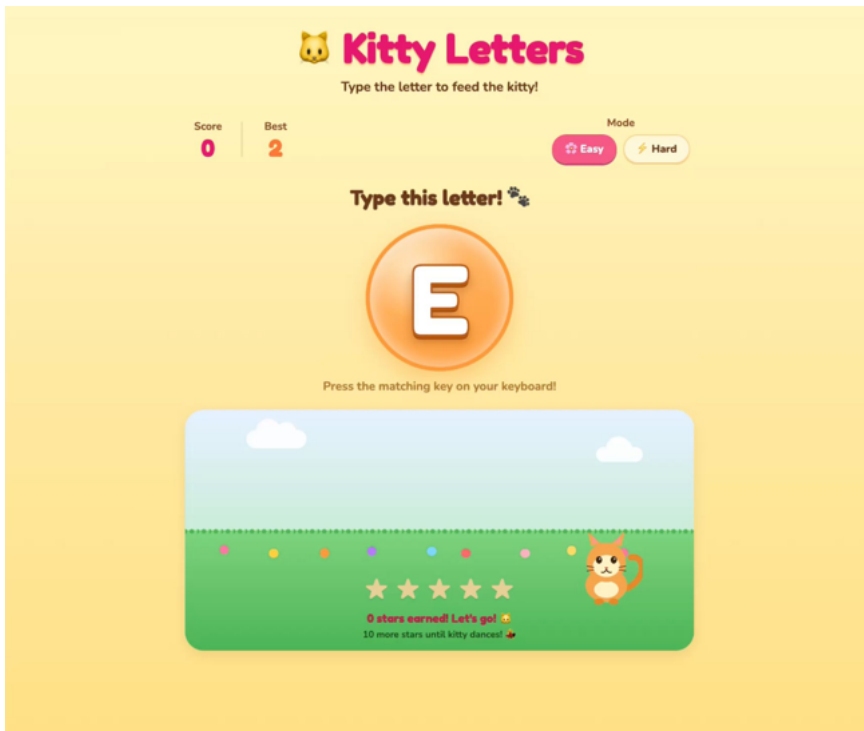
Generated with Seedance 2.0

AI models can generate images, videos, voices, music, code, and more

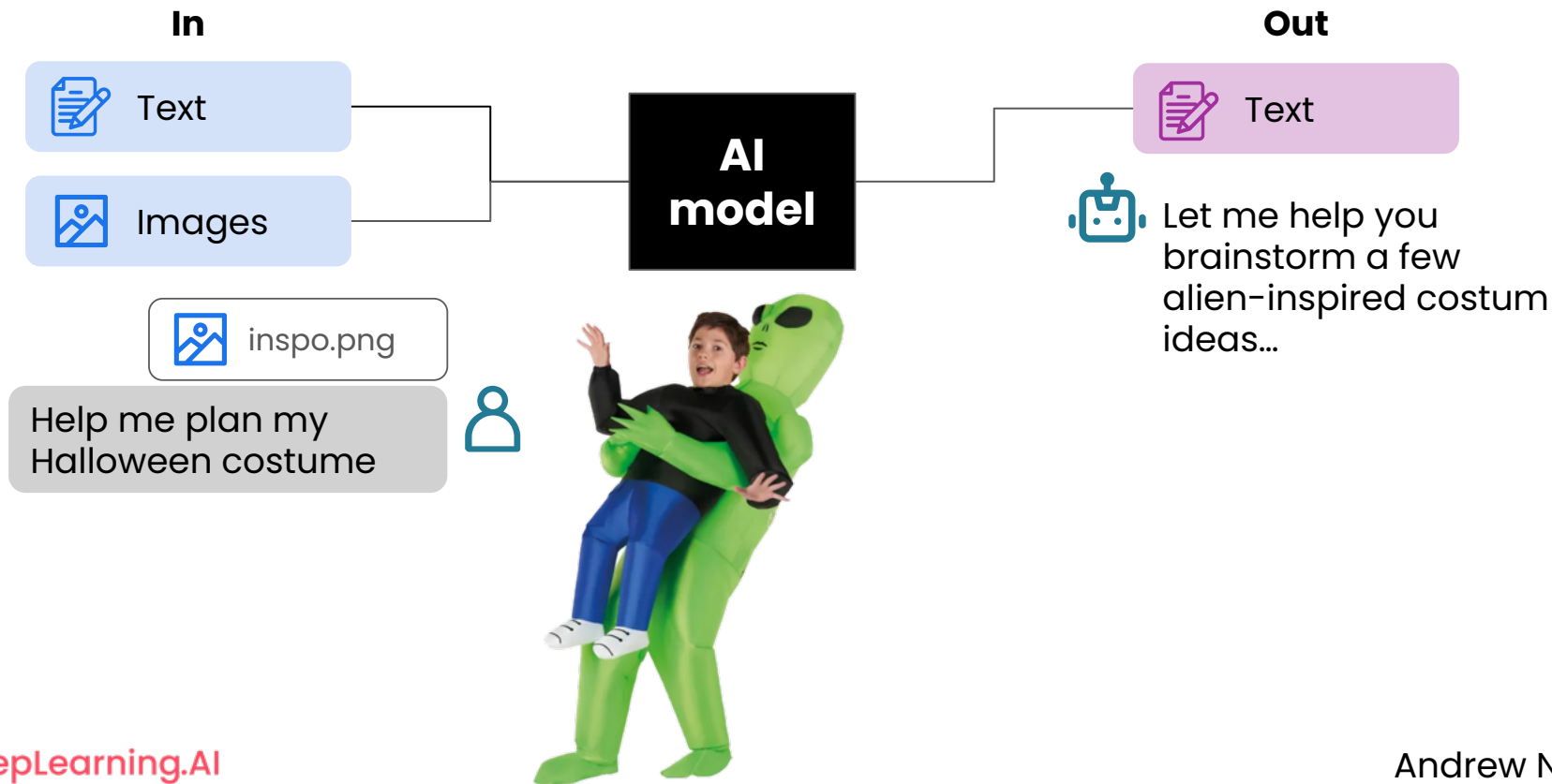


Voice clone

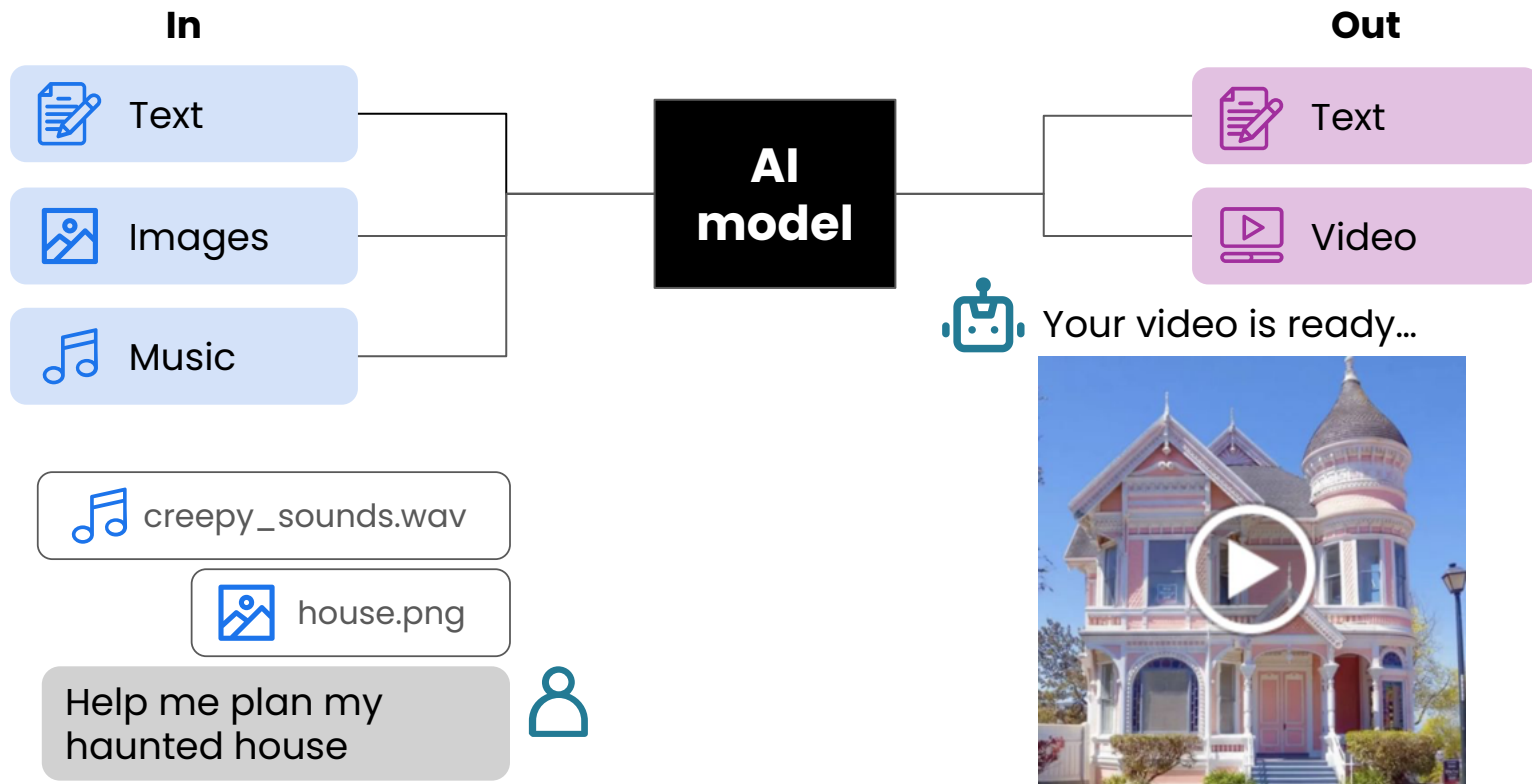
A cat-based typing game



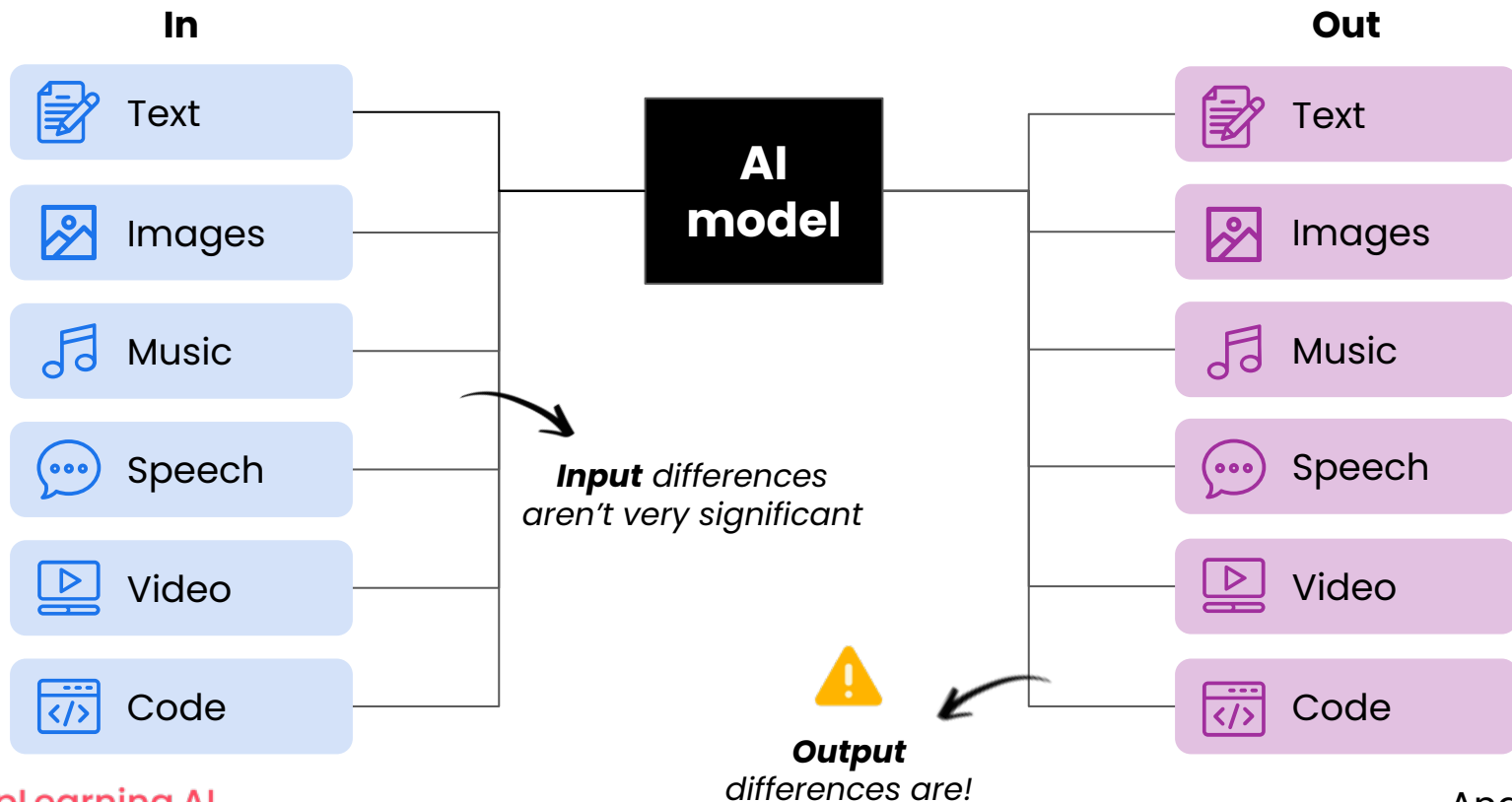
There are many combinations of input & output



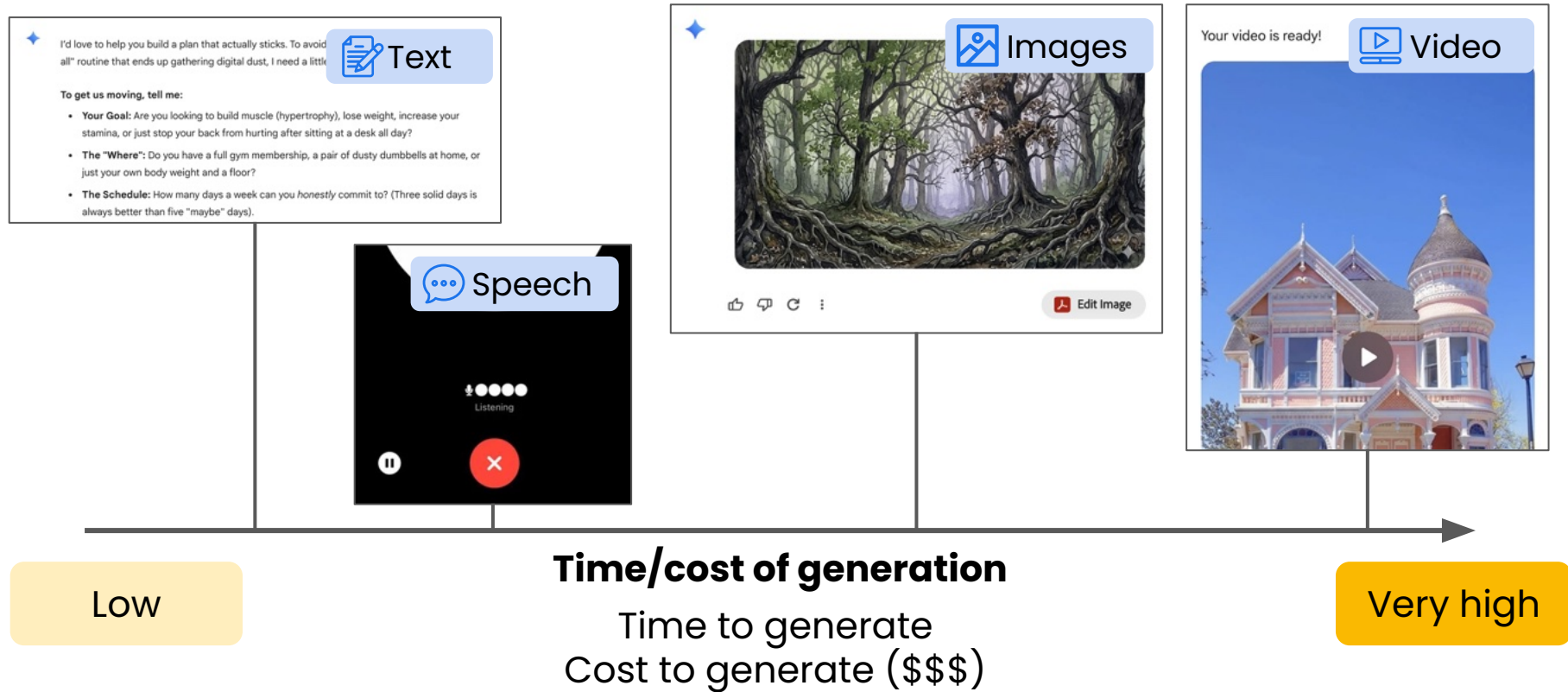
There are many combinations of input & output



There are many combinations of input & output



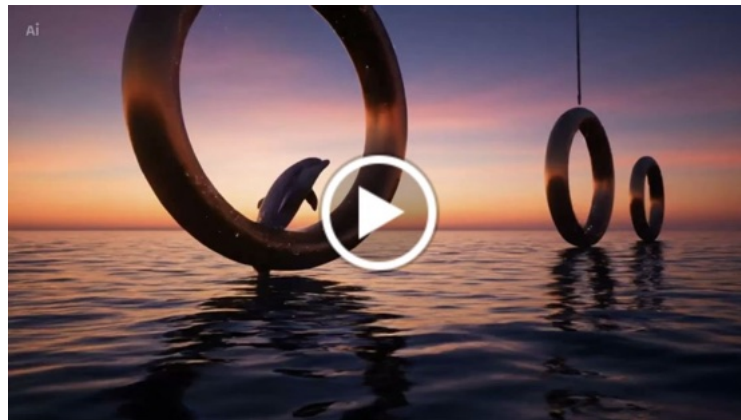
Some data types are slower & more costly to generate



Multimedia generation has progressed significantly



Generated with imagen Video (2022)



Generated with Seedance



Old AI-generated voice



Recent AI-generated voice

Multimodal data: most prompting principles still apply!

Prompting techniques are affected by how slow & costly multimodal generation is



Easier to apply

- More context is better
- Use the best model available

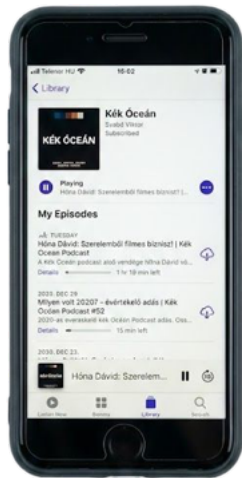


More difficult to apply

- Generate multiple options
- Iterate

With great power comes great responsibilities

AI technologies can be used for good or for harm, like voice generation:



Fixing mistakes
in a podcast



Giving characters
lifelike voices



Scamming someone
with a relative's voice

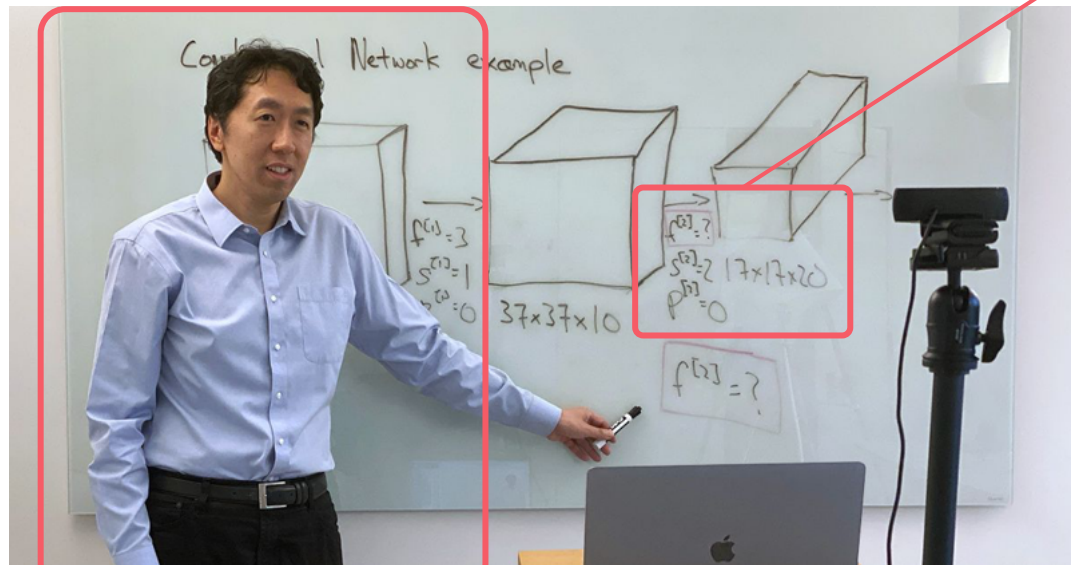


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Image understanding

Using images in prompts



Complicated
math concepts

What is this class about?



He's teaching a
convolutional neural network:

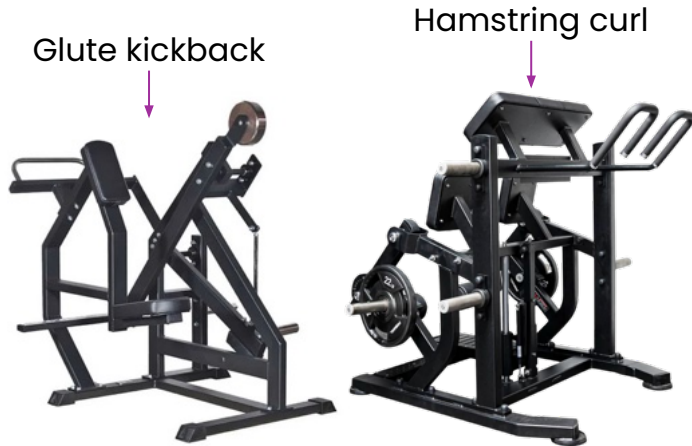
- 37x37x10 input volume

...

The question mark on $f[2]$
suggests he'll have students
compute that...

Professor blocking
whiteboard

Fine-grained details may be missed



What are these machines at my gym?



Left: chest press

Right: ab crunch



Very
visually
similar!



Create a sales ad for this item



[FOR SALE] Human-sized
hamster wheel treadmill...

Visually distinct object

Using images with text



What's my portion of the bill?
I had: taro milk tea, vegan ramen,
half of the gyoza.

Your part of the bill is \$27.30:

Ramen	\$16.95
½ gyoza	\$3.25
Taro milk tea	\$5.70
Tax	\$1.88
Your total	\$27.78

Try it!

Can you read this journal entry?



The careless beat of minute waves [against] the rocks I sit on, make me increasingly drowsy and relaxed. I saw my dreams today...

topped by an almost perfectly symmetrical
leaves. The careless beat of minute waves
the rocks I sit on, make me increasin
over, and ^{relaxed} I saw my dreams today...

ling hills, covered with small yellow
lowers, cut off abruptly, as cliffs in
the sea. I am in awe. Looking off
to the distance... snow capped mo
to the sky, beautifully rolling but smoothly
orned clouds. Vaguely appearing like
ondora in ~~the~~ flight. The sun
still out, and is beginning to set. The
now on the Orono Volcano turns orange, and
see the sun on the opposite horizon. As

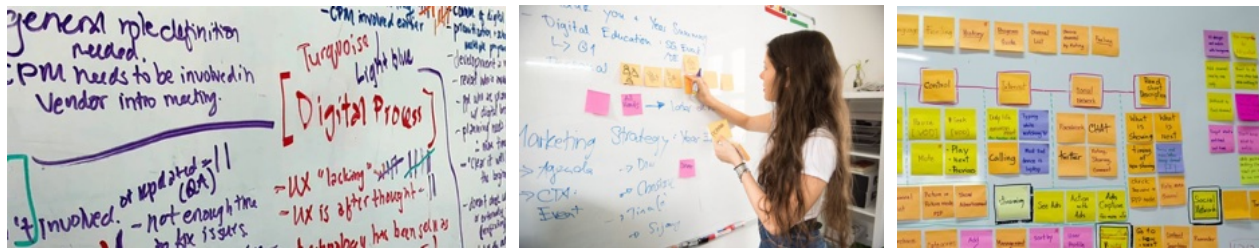
Build an archive of my family's history
based on these handwritten letters



Adding many images to a prompt



notes.docx



Summarize our ideas from today's brainstorming meeting



Meeting Summary – Brainstorming Session

The top 3 ideas that came up were...

Recap of tips for prompting with images

1

AI can read basic text in images

2

Visual understanding may miss details

3

Use many images when needed





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Working with Multimedia & Code

Image generation

Image generators can edit images



Remove the glare and the rough texture underneath. And make it a more natural aspect ratio



Sure! Here is your restored photo.



Developing image generation prompts

Generate a prompt for an image of a cat secretly running a coffee shop at night



Whimsical nighttime café run by a clever orange cat wearing a tiny apron, brewing espresso behind the counter, cozy café interior, cartoon style...

Setting

Character details

Mood/style



The language of images

Art & art history buffs excel
at image prompting!

Cinematic



Watercolor



Cyberpunk



Anime



How image generation works

- Text models: produce output piece by piece
- Image models: generate the image all at once

Done by a
diffusion model

Training

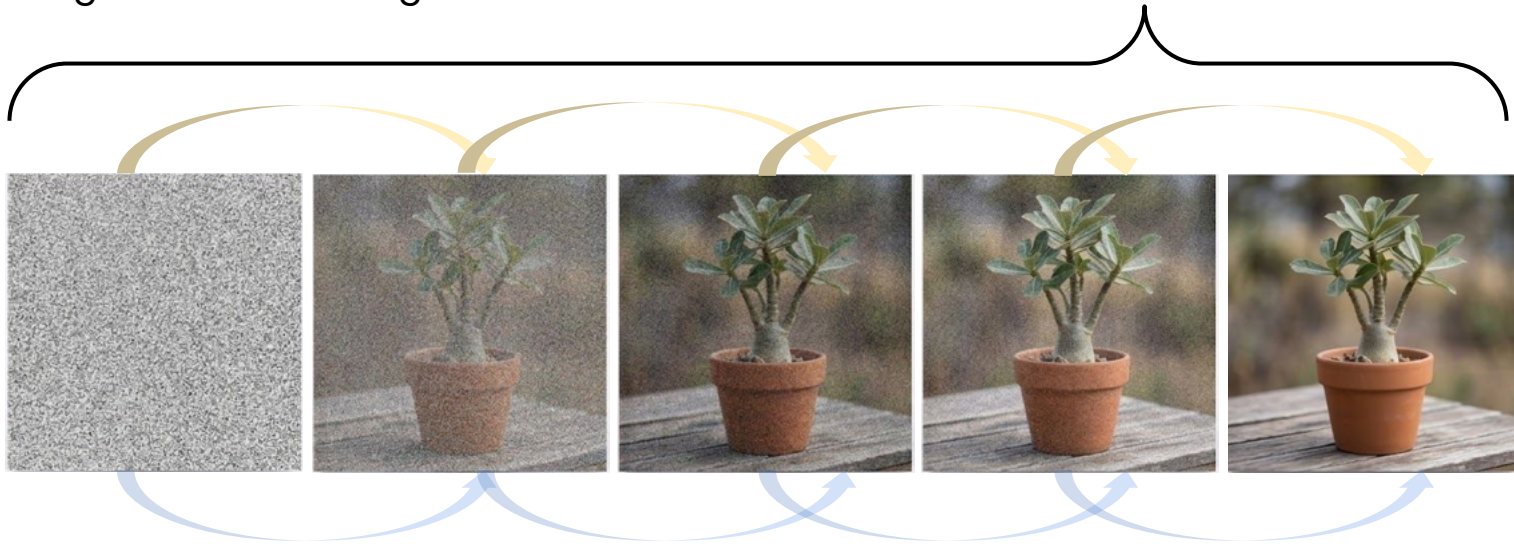
Caption

Small **potted plant** on a wooden **table**

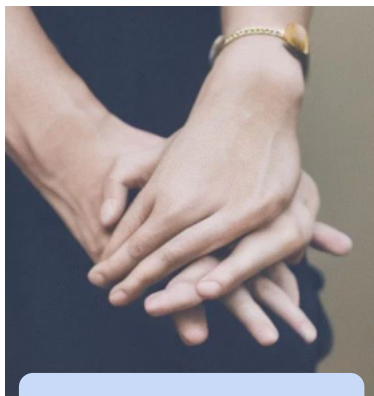
Prompting



Create an image of a
potted plant on a **table**



Diffusion models generate random outputs and have errors



Weird hands

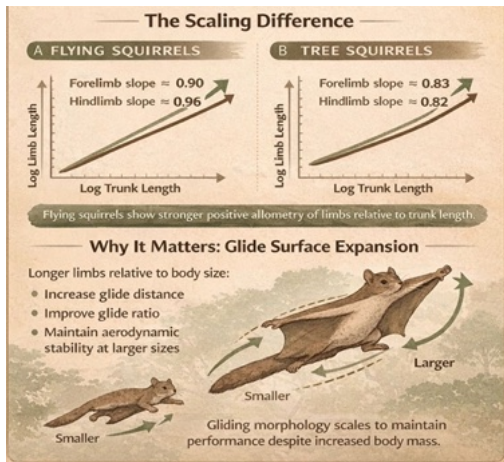


Garbled text



Inconsistent characters

Generate an infographic of the key points in this paper



BEHOLD: THE FLYING SQUIRREL SPACE SUIT!

FLYING SQUIRREL SPACE SUIT

- HELMET (ACORN CLEAR VISION)
- GLOVE WINGS (DEEP SPACE AEROMEMBRANE)
- THRUSTER BOOSTERS
- TAIL PORT (FOR MAX BALANCE)
- POCKET FOR SNACKS

SUIT UP!

READY FOR ANY GALAXY!

LAUNCH!

GLIDE. EXPLORE. COLLECT ACORNS.

THE UNIVERSE IS FULL OF TREES I HAVEN'T MET YET!

Image generation can be slow & costly

	Speed	Cost	Delivery
Text response	Seconds	Less than \$0.01	Piece by piece, can stop early
One image	10s of seconds	Many cents	All at once, can't stop early

Why image generation is often limited!

♦ I'd love to help you build a plan that actually sticks. To avoid giving you a generic "one-size-fits-all" routine that ends up gathering digital dust, I need a little bit of intel on your starting line.

To get us moving, tell me:

- **Your Goal:** Are you looking to build muscle (hypertrophy), lose weight, increase your stamina, or just stop your back from hurting after sitting at a desk all day?
- **The "Where":** Do you have a full gym membership, a pair of dusty dumbbells at home, or just your own body weight and a floor?
- **The Schedule:** How many days a week can you honestly commit to? (Three solid days is always better than five "maybe" days).





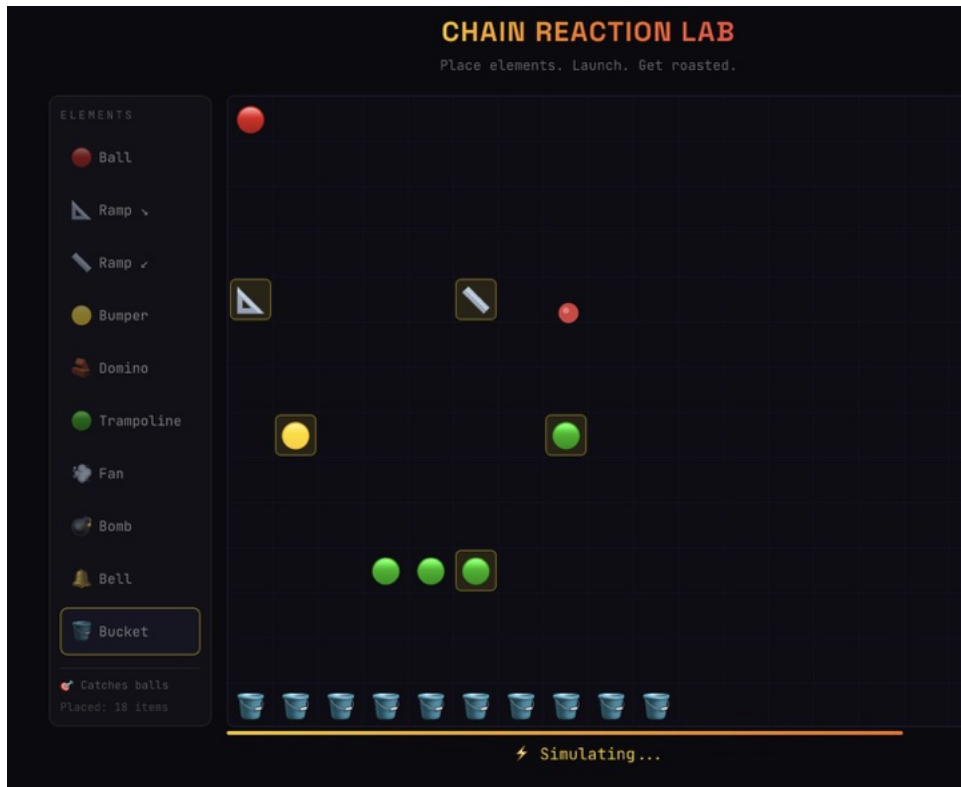
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Building apps

Chain reaction game

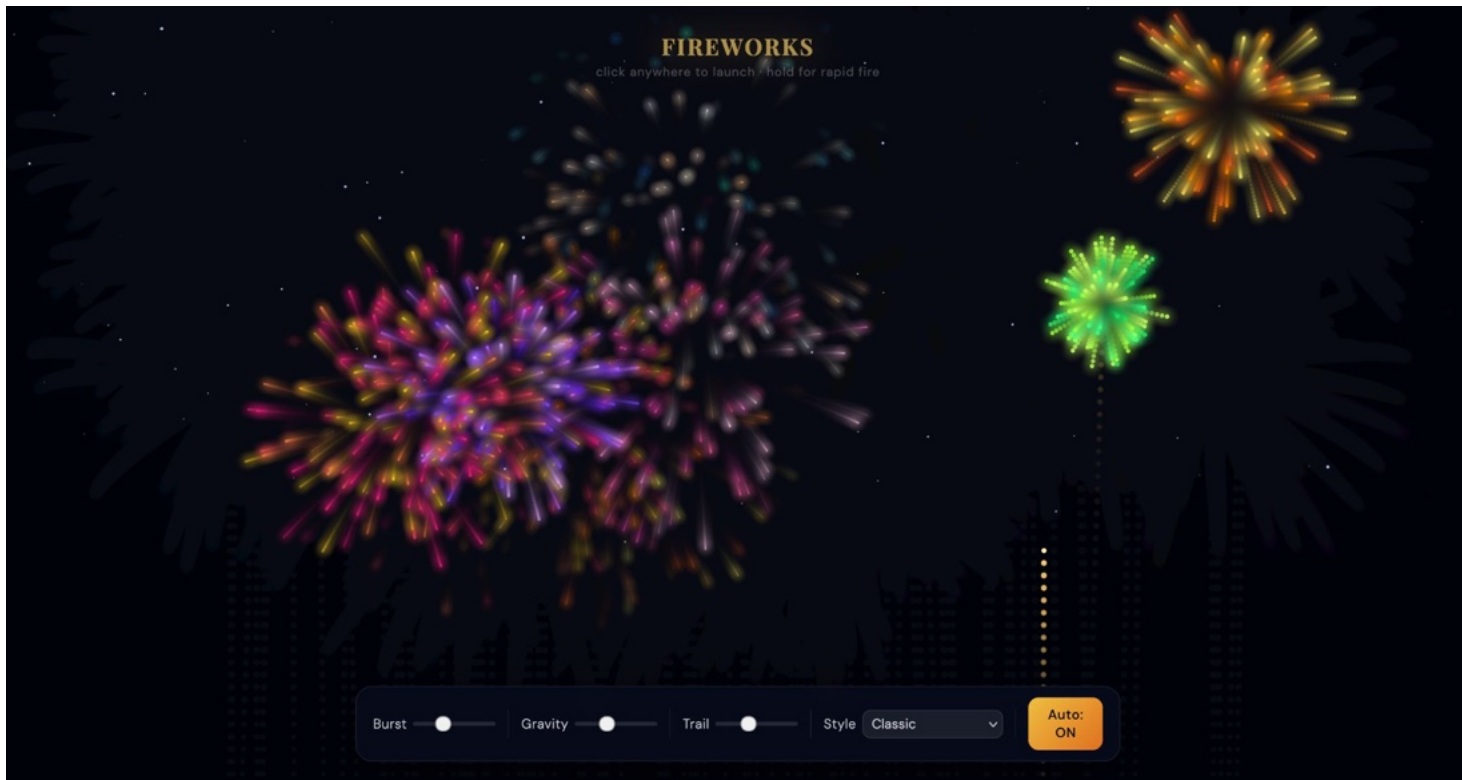
Build a game that involves placing obstacles and a goal and produces a simulation of what I designed.



Andrew Ng

Fireworks simulator

Build an interactive fireworks app.
When I click, launch a firework from
the bottom and explode it

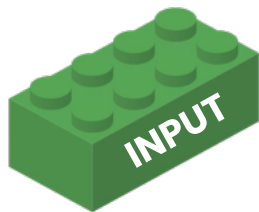


Andrew Ng

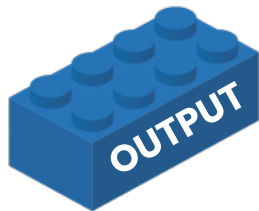
Use these building blocks to prompt your app



What to
create



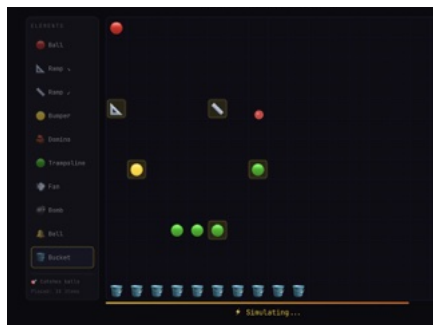
What users
provide



What the
app does



Generate a fun fireworks simulator. I should be able to click on the screen and see a colorful display of fireworks.

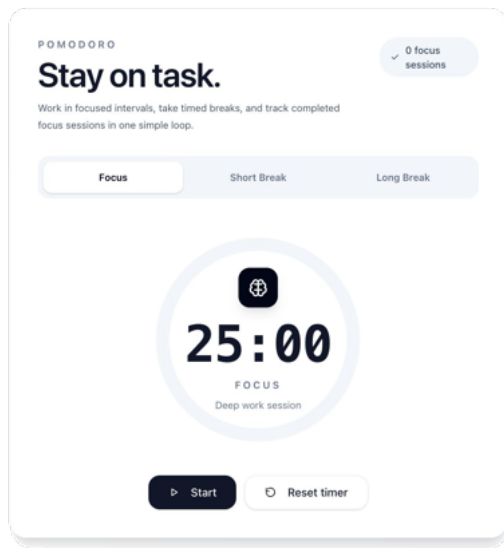


Generate a fun game that involves placing obstacles and a goal and produces a simulation of what I designed.



Apps can also be functional, not just fun!

Pomodoro timer



POMODORO

Stay on task.

0 focus sessions

Work in focused intervals, take timed breaks, and track completed focus sessions in one simple loop.

Focus Short Break Long Break

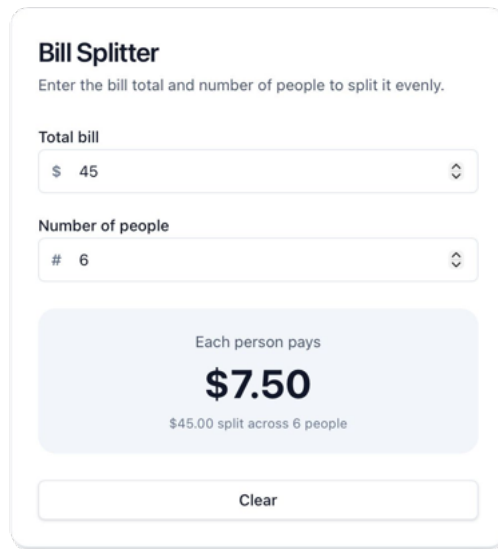
25:00

FOCUS

Deep work session

Start Reset timer

Bill calculator



Bill Splitter

Enter the bill total and number of people to split it evenly.

Total bill

\$ 45

Number of people

6

Each person pays

\$7.50

\$45.00 split across 6 people

Clear

Outfit picker



what to wear

14

that's 57°F

feels lovely

SWCS

SUNNY CLOUDY RAINY SNOWY WINDY

HEAD no hat needed

TORSO rain shell over cardigan or light hoodie

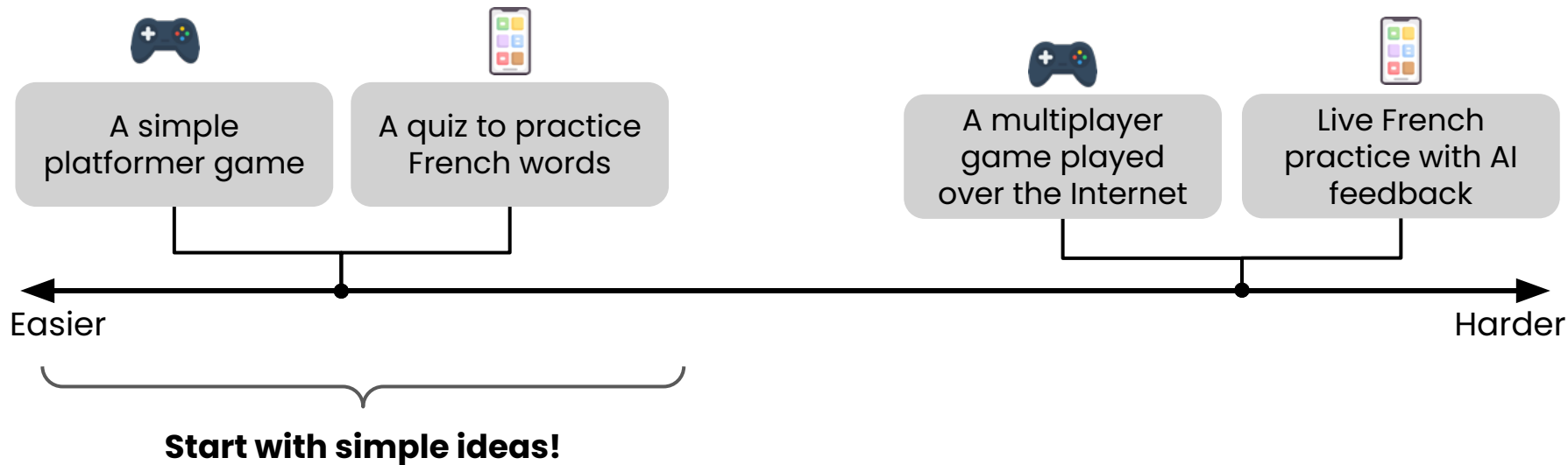
LEGS pants or relaxed chinos

SHOES waterproof shoes or rain boots

EXTRA umbrella, ready to deploy

Experiment with simple app ideas

Some ideas are easier to create than others.





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Data analysis

Analyzing your personal data

How are my pace and distance progressing?



Date	Activity	Km	Min/km	Notes
2/10/2026	Run/Walk	1.8	8:10	First run, lots of walking intervals.
2/12/2026	Run	2	7:45	Felt better than expected.
2/15/2026	Run/Walk	1.5	8:30	Struggled today, high humidity.
2/18/2026	Run	2.5	7:51	Pushed to a new distance goal.
2/21/2026	Rest	0	-	Busy weekend, missed run.
2/23/2026	Run	2.2	7:44	Back on track, focused on pace.
2/25/2026	Run	2.8	7:43	Found a better route.
2/27/2026	Run	3	7:55	First time hitting 3km!
3/1/2026	Run	3.2	7:48	Longest run yet.
3/3/2026	Run	3	7:35	Focus on consistency.
3/5/2026	Run	3.5	7:30	Sore knees, took an extra loop.
3/7/2026	Run	3.5	7:30	Consistent Monday
3/9/2026	Run	3.1	7:42	

Running tracker data



Analyzing...



AI models can write and run code

Date	Orders	AvgOrderValue (\$)	Revenue (\$)
2025-02-09	45	55.20	2484
2025-02-10	41	54.88	2250.08
2025-02-11	42	55.24	2320.08
2025-02-12	43	58.14	2500.02
2025-02-13	47	60.11	2825.17
2025-02-14	55	70.33	3868.15
2025-02-15	60	57.00	3420
2025-02-16	45	55.48	2496.6
2025-02-17	42	59.89	2515.38
2025-02-18	50	54.90	2745.6
2025-02-19	48	52.83	2538
2025-02-20	59	61.18	3609.62

Small business sales data

What can you tell me about this month's sales?



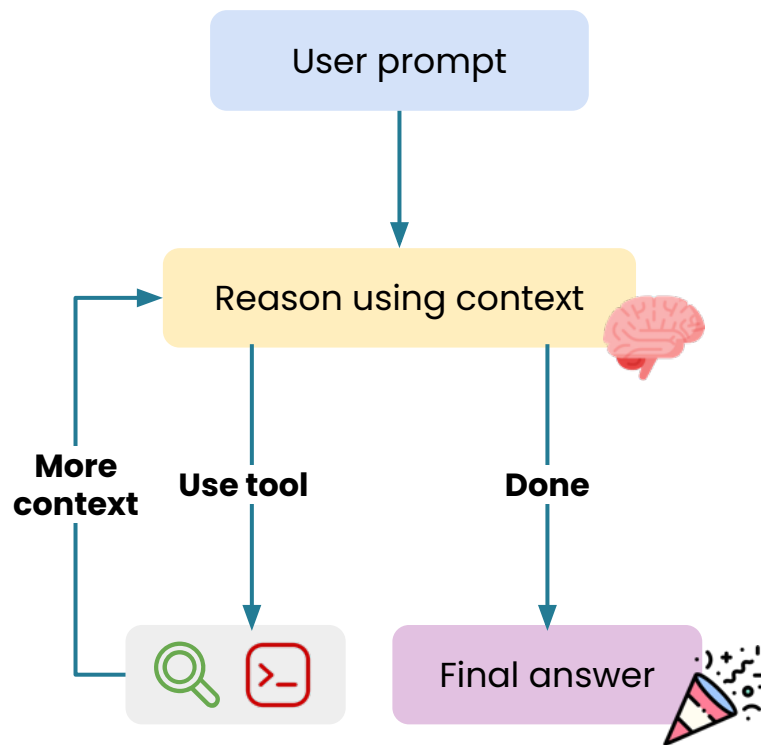
Analyzing...

Calculating monthly revenue...

Creating a graph...

I analyzed trend, seasonality, promos, and outliers...

Running code is like any other tool



Getting insights from data



sales.xlsx



Inspecting the data...



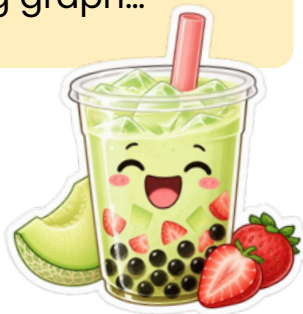
Calculating month-by-month changes...



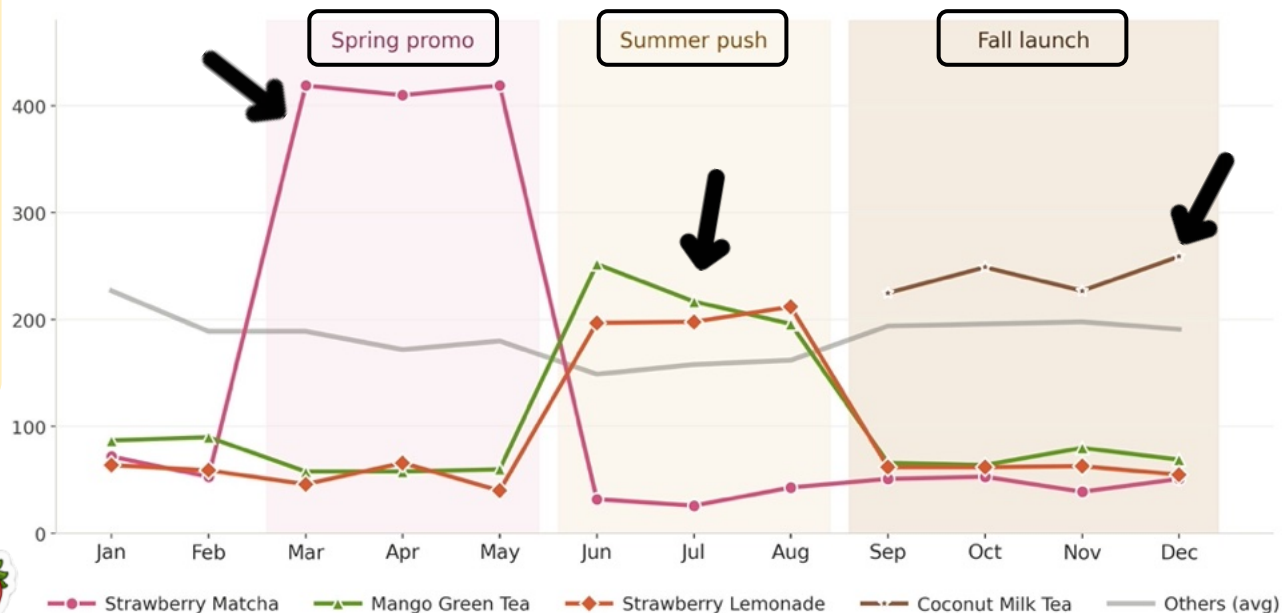
I'm noticing some clear patterns. Most drinks are flat, but four stand out. I'll graph those.



Generating graph...



Which drinks had the biggest changes in sales? Graph it.



Getting insights from data



Create a one-slide year-in-review graphic for our bubble tea shop. **Analyze the data carefully** for insights.



Andrew Ng

When AI chooses to code

Question	Source	Reason
Why do cats stare at walls?		Common knowledge on the Internet
Find me a highly rated gym nearby		Location-specific
Build me a plan for a haunted house		Synthesis needed
What are my sales trends?		Calculation & graphing needed